

QUALITY AND REGULATORY SOLUTIONS

for PAT in Continuous Manufacturing

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Process analytical technology (PAT) is perceived as the main enabler for a robust control strategy with continuous manufacturing (CM) because PAT can aid in implementing CM throughout the entire life cycle. This article discusses quality and regulatory hurdles in the life cycle of a PAT application—including model life-cycle management—in combination with CM for small and large molecules, with the goal of proposing strategies to resolve each challenge.

Continuous manufacturing represents the next generation of pharmaceutical manufacturing processes for both large and small molecules. It is recognized by regulatory authorities as a key emerging technology. The US FDA has approved various small molecule products and recently issued draft guidance for industry on CM [1, 2]. In this article, we do not focus on the capabilities of specific PAT technologies, which have been reviewed in other recent publications [3–6]. Instead, we address how companies can add flexibility and maximize the value of PAT for CM. As we discuss general issues, we focus on how they apply to all or most PAT tools and we use some common PAT tools as aids to describe example scenarios.

Important regulatory guidance on PAT includes ICH Q8 (R2) [7] and the FDA's "Guidance for Industry" from 2004 [8]. The FDA PAT guidance considers PAT to be,

A system for designing, analyzing, and controlling manufacturing through timely measurements (i.e., during processing) of critical quality and performance attributes of raw and in-process materials and processes, with the goal of ensuring final product quality. It is important to note that the term analytical in PAT is viewed broadly to include chemical, physical, microbiological, mathematical, and risk analysis conducted in an integrated manner."

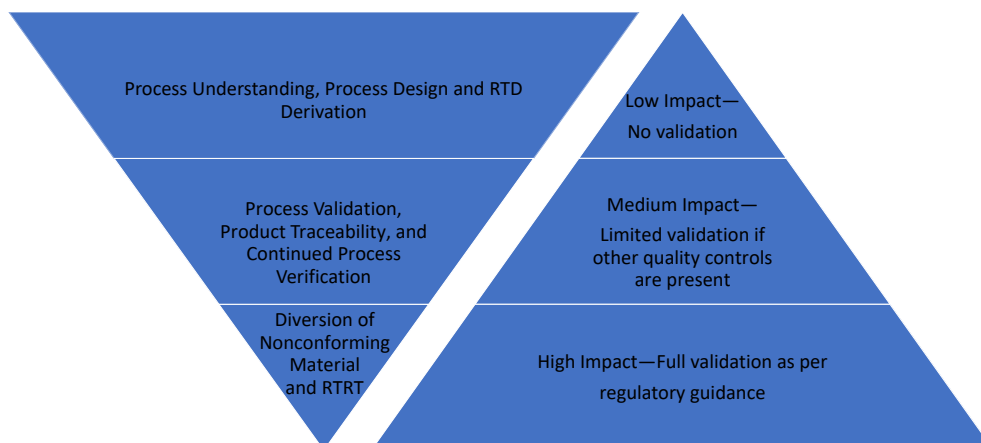
For CM, analytics must move closer to the process. PAT can therefore add significant value in design, analysis, and control of the CM process. Because PAT does not involve pooling or holding of the process, decisions can be made in real time. For example, when designing equipment or modeling unit operation behavior, PAT measurements used to determine residence time distribution (RTD) are critical. During process development, PAT may be used to confirm satisfactory process operation, verify models, inform on divert to waste (DTW) situations, or perform feedback control. The control strategy for a continuous process may also benefit from the use of PAT. Furthermore, PAT can be used for final critical quality attribute (CQA) determination directly or as input to a more complex model that includes process parameters and material attributes.

When developing and implementing a commercial CM line, PAT is recommended, even more strongly than it is for batch manufacturing, for all three phases (design, analysis, and control). However, the extent of PAT use is company- and product-dependent, reflecting the perceived return on investment.

On the continuum from simplest to most complex applications, the optimum level of PAT deployment in various stages of the CM life cycle, from design to routine operation, is a company decision that may be based on perceived risks associated with regulatory and quality considerations. The validation of PAT applications must evolve during their life cycle to ensure they remain fit for purpose. Lack of clarity on regulatory expectations at different points in the life cycle can derail PAT early in the process.

Figure 1 shows how the number of PAT applications tends to decline as we move from process understanding to real-time release testing (RTRT) (left triangle), whereas validation requirements (right triangle) increase as we move from low-impact to high-impact applications. Because successful implementation of PAT can add benefits to a CM process, we suggest in this article ways to increase the utilization of PAT within all stages of the left triangle.

Figure 1: Conceptual representation of the number of pharmaceutical industry–deployed PAT applications in use contrasted with the resources required to develop, validate, support, and maintain the methods and enable defined fit-for-purpose testing.



PAT IN DEVELOPMENT

During early process development, having the tools in place for rapid analysis and visualization helps turn PAT data into immediately actionable knowledge. At this stage, companies often seek to enhance understanding to effectively implement a CM platform, and it might be easy to install a preliminary PAT system for this purpose. Notably, these early data could form the basis of differentiation between good and bad product that determines quality decisions. For this reason, a close-to-GMP-ready system to manage the PAT may be needed earlier in the development process than would be required for a batch process; also, a plan on how to leverage the preliminary system may be required if a permanent system is needed for commercial use. However, in most scenarios, companies do not need to install a commercial release-ready GMP system for initial process development.

Data capture with a PAT system should meet process-specific criteria that are based on the targeted CQAs according to quality by design (QbD) principles. A potential strategy would be to start by using PAT tools, such as semiquantitative or fingerprint methods, to detect relative changes [9]. These are easier methods to qualify or validate, which would then allow organizations to use PAT data for quality decisions and help companies leverage the data for future, more advanced controls or decisions. This approach could also streamline the requirements around robustness and system suitability, which may require significant resources in early development. It would be beneficial if a company established a process for qualified versus validated methods similar to what has been done for traditional testing.

To clarify the relationship between RTD and PAT, the FDA published the initial PAT framework [8], which supports the move from static batch processing to more-dynamic approaches that mitigate the risk of producing poor-quality product. ICH implemented a trio of quality guidance: Q8(R2), Q9, and Q10 [7, 10, 11].

Continuous systems with automation and process control frequently result in high-quality (low-variability) products, whereas traditional batch processing can be less understood, resulting in less-predictable product quality [12].

Using high-frequency PAT sensors can ensure adequate sensing to determine the approximate RTD shape. Aided by a simple peak detection algorithm, analysts can easily detect the most significant spikes. It is logical to assume that detecting the downstream response is easier than detecting the pulse disturbance itself. If a plug flow process were expected, the perturbation would be largely unchanged as it traveled along the system, meaning pulse into the system would result in a pulse response, although this would be difficult to detect without a very rapid measurement. RTD plays an important role in material traceability because it characterizes the spreading of the materials through the system. Thus, a disturbance could be predictively tracked through the entire continuous system, enabling downstream control or even removal of the affected material.

The determination of RTDs is one of the primary areas in which PAT is used in development. RTD aids in the traceability of raw materials in the manufacturing process, which is critical when material is continuously fed into and removed from a process. Once the RTD has been determined for each unit operation, as well as the integrated line, it can be used to facilitate material traceability and help determine sensor placement and measurement frequency to ensure that any unacceptable material variations can be detected, recorded, and addressed.

AUTOMATION AND RTD

Continuous systems with automation and process control result in high-quality, low-variability products. A properly designed continuous system with appropriate sensors to record and display process parameters, such as critical process parameters (CPPs) and

Figure 2: A conceptual integrated small molecule CM process.

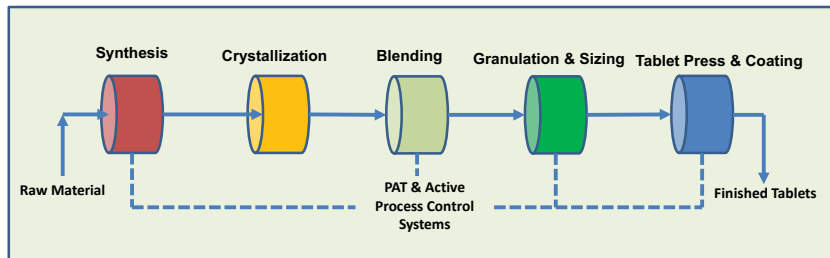
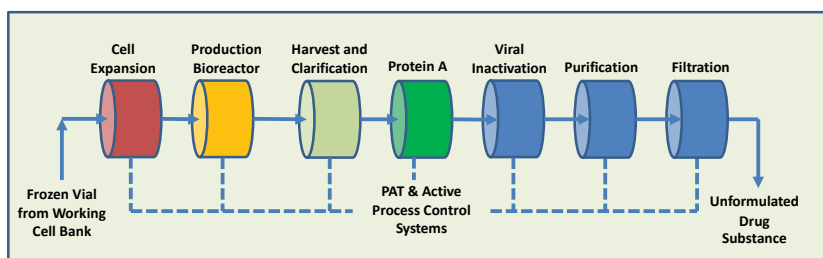


Figure 3: A conceptual integrated large molecule CM process.



non-CPPs, handles small portions of material at any given moment, increasing material monitoring scrutiny. When CM uses product and process knowledge with properly implemented online PAT, it can meet the criteria required to enable RTRT, leading to rapid and reliable batch release of high-quality product (see Figures 2 and 3 for CM processes for small and large molecules, respectively). Despite these vast advantages, CM also has significant challenges, and continued focus by industry and regulators is required to resolve them.

In the chemical processing field, the RTD is used to describe how a material travels inside the unit operations of a continuous process system. RTD is a critical tool in pharmaceutical process understanding, quality assurance, and equipment and sensing design, but it is underutilized due to a lack of acceptance of RTD as an alternative to testing, especially if an automated RTD control system is available as part of the line.

As mentioned, PAT can be a useful tool as part of the control strategy, which is generally developed and initially implemented for production of clinical trial materials. The control strategy is usually refined for use in commercial manufacture as new knowledge is gained. Changes could involve acceptance criteria, analytical methodology (including switching from traditional lab testing to PAT or model-based methods), or the points of control (e.g., introduction of RTRT). Stakeholders across an organization must agree whether the documentation and equipment qualification of any PAT tool done during development activities are sufficient to permit the leveraging of the development data into a final RTRT method that incorporates increased process knowledge and

finalized process controls. Additional emphasis on process controls should be considered in cases where products cannot be well characterized or quality attributes might not be readily measurable due to limitations of testing or detectability (e.g., microbial load/sterility, low drug load).

PAT VALIDATION AND LIFE-CYCLE MANAGEMENT

As noted previously, PAT benefits a CM line by helping answer technical and quality-related questions about product and process performance, and by providing the direct answer to product quality requirements. As mentioned, the main hurdles of validation strategy of the PAT and the accompanying methods arise during development. The qualification and validation of PAT applications have significant overlap with traditional analytical equipment qualification and validation and should be conducted in accordance with ICH Q2 (R1) [13], where the capabilities of the equipment must align with the type of test that the technology will address (ID, qualitative, or quantitative measure). When PAT is incorporated into the manufacturing process as inline or online tools, additional issues such as probe fouling and complexity of CM cleaning validation should be addressed. Some examples of how to incorporate these aspects into the line risk analysis are described in detail in ASTM standard E2898-14 [14]. Figure 4 shows the life cycle for method development and validation of analytical methods for PAT applications that is aligned with E2898-14.

E2898-14 includes paths for PAT method validation, including how a risk assessment can document a risk-based approach that identifies what parameters to address based on the use of the PAT

tool and how each parameter should be handled. (This risk-based approach can be found in ASTM E2476 [15].)

Examples of potential areas for confusion around validation requirements in the pharmaceutical industry include:

- How to translate the risk-based approach for a scenario in which PAT is used for DTW and not part of the final product release strategy
- How to validate the PAT measurement if equivalency to a traditional method is not feasible or no traditional methodology is available
- How to leverage development work or data generation when the model is of a lower impact level (Table 1)

Internal education of employees regarding the similarities and differences between PAT and traditional testing is essential to successfully implement and sustain PAT as a tool for ensuring quality. Training programs are needed for all levels of the organization; the scope of education will range from awareness to detailed training in how to install and maintain the PAT, including interfaces with the manufacturing line.

Many common PAT tools require the use of chemometric models to generate the required data. At any stage during the life cycle of the application (e.g., feasibility, development, validation, or routine use for product manufacture), the level of validation of the model, and thereby the validation of the test method, must be

aligned with the impact of the application at that time (Figure 1). Table 1 categorizes the impact of the model use into three levels [16] and presents examples and the proposed level of validation/documentation appropriate for the situations.

Table 1 captures the current information from the FDA guidance document [16] and shows that the regulatory expectation on validation and documentation increases as the potential impact of the model on quality increases. The information in Table 1 and the

Figure 4: General flow for the PAT method life cycle.

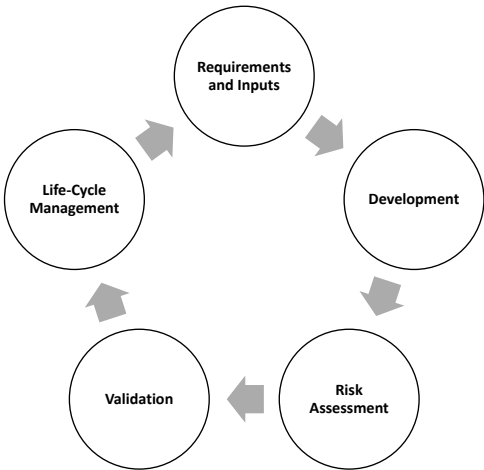
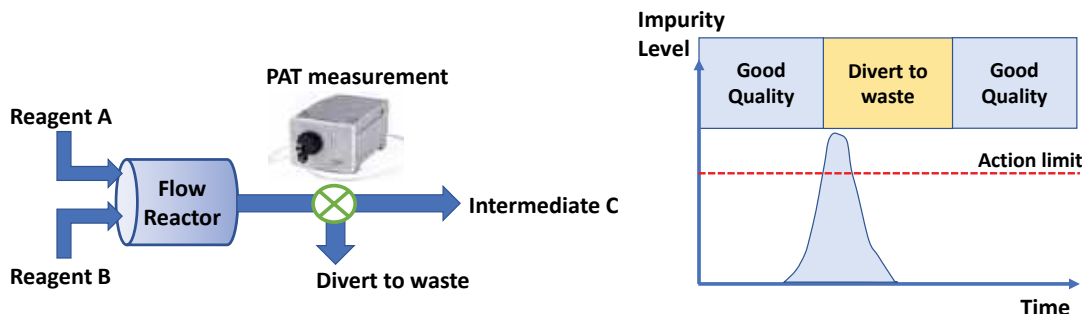


Table 1. Model validation and documentation expectations.

Category	Impact Level		
	Low	Medium	High
Model use	To support product or process development	To help ensure product quality (model predictions are not the sole indicator of product quality)	To predict product quality (model predictions are significant indicators of product quality)
Examples	Formulation, process optimization, and scaling	Design space models, in-process control models; PAT for process control when not part of control strategy	Chemometric models for product assay, surrogate model for dissolution PAT for process control when part of registered control strategy
Validation	Formal validation is unlikely Method development will not include all variability expected in commercial manufacturing	Limited validation may be appropriate	Full model validation is done following relevant guidelines
Documentation	Includes discussion of how the models were developed and used to guide process development	Model assumptions; a tabular or graphical summary of model inputs and outputs; relevant model equations (e.g., for mechanistic models) either in the submission or via a reference; statistical analysis where appropriate; a comparison of model predictions with measured data; and a discussion of how the other elements in the control strategy help mitigate uncertainty in the model, if appropriate	Data and prior knowledge (e.g., for established first principles-driven models), such as model assumptions; appropriateness of the sample size, number, and distribution of samples; data pretreatment; justification for variable selection; model inputs and outputs; model equations; statistical analysis of data showing fit and prediction ability; rationale for setting of model acceptance criteria; model validation (internal and external); and a general discussion of approaches for model verification during the life cycle

Figure 5: Illustration of a medium-impact PAT application applied to flow chemistry. Demonstration of the concept of PAT measurement prior to a DTW valve (left) is not directly linked in scale to the graph (right).



FDA guidance [16] can be leveraged to create appropriate internal requirements depending on the intended use of a model at any phase of the PAT life cycle. It should be noted that based on long-term intent of a model, a company can decide to increase the validation and documentation requirements at an earlier phase of application development to facilitate the end use. In a similar manner, qualification activities may be performed earlier for chemical manufacturing control (CMC) deliverables.

Regulatory guidance documents for near-infrared spectroscopy (NIRS) cover what is in or out of scope. For example, European Medicines Agency (EMA) NIRS guidance [17] states that “NIRS for non-regulatory purposes, such as generating process knowledge, is out of scope of this guideline.” FDA guidance on NIRS [18] applies to applications used “during the manufacture and analysis of pharmaceuticals (including raw materials, in-process materials and intermediates, and finished products).” It is generally assumed that this FDA guidance applies to applications that will be included in regulatory submissions aligned with EMA NIRS guidance, although that is not specifically stated. The FDA and EMA NIRS guidance can be applied to similar PAT technologies.

According to current guidance documents, organizations must obtain additional regulatory approval during the life cycle of the PAT application. This is a challenge, and it is compounded by the differences in the requirements of various regulatory bodies. Certain changes to the scope of the NIRS method (EMA guidance) require regulatory notification and acceptance before a model update can be implemented [19]. FDA guidance also requires a prior approval supplement when there are major changes to the application or a change-being-effected in 30 days (CBE-30) filing for a moderate change. If the PAT application were required to manufacture using continuous processing, manufacture would have to cease during this time period unless an alternative control strategy that did not rely on the PAT application were accepted by regulatory authorities.

The following are two potential solutions to mitigate this challenge in EMA guidance and general expectations of approval of model changes:

- The use of performance-based established conditions (ECs) for PAT methods versus parameter-based ECs: This option would require a clear description of model outputs that must be met, such as accuracy, precision, linearity, and so on, instead of just the software version, spectral pretreatment, and equipment model. If performance-based ECs were used, a range of changes to the PAT method would not require a filing because those changes would be verified through the confirmation of the output and tracked within the pharmaceutical quality system (PQS).
- The use of postapproval change management protocol (PACMP): This is a two-step process. First, the organization submits a written protocol that describes the proposed change, its rationale, and risk management, as well as proposed studies and acceptance criteria to assess the impact of the change(s). Regulatory authorities must approve this protocol in advance of the implementation of the proposed change(s). Second, the organization submits the actual results/data based on the protocol approved by the regulatory authorities and according to the agreed categorization (classification). In certain cases (e.g., noncritical or repetitive changes), regulatory approval of the second step may not be required because that step will be managed within the applicant’s PQS.

The second approach can result in the downgrade of notification requirements (e.g., 1B to 1A_{IN}), which could obviate the need to halt CM while awaiting regulatory approval. Overall, this is an area of ongoing conversation between industry and regulators.

The goal of PAT model validation is to have a fit-for-purpose validation and documentation approach for model use. In general, industry stakeholders agree on health authority expectations for low- and high-impact models. The requirements for medium-impact models are less clear.

Consider the following example, in which two reagents (A and B) are reacted together in a single-stage flow process to form an intermediate (C) of a multistage synthetic process (Figure 5). A PAT

measurement is made at the exit of the reactor, which is capable of quantifying an impurity in solution that is difficult to purge and can impact API quality downstream. Intermediate (C) is collected in a batch vessel, sampled for analysis, and tested against a specification, irrespective of the PAT measurement. This offline analysis forms the basis for ensuring control of the impurity. However, using PAT to DTW at the appropriate times can reduce the level of impurity in the intermediate or potentially prevent a batch failure. (With enhanced confidence in the PAT method and RTDs, the extended DTW time shown in Figure 5 may be reduced to match the point where the impurity is below the action limit.)

In this case, the application should be considered medium impact. The level of assurance required for this PAT measurement is less than if the impurity were not controlled by subsequent downstream analysis, and the process can run without the PAT measurement. Fit-for-purpose application validation is required, as this is a business risk and presents no risk to the patient.

Similarly, consider a PAT measurement performed on a mixture or blend for an oral solid dose formulation used to ensure proper composition to determine when to DTW. If the final product has another PAT or traditional test with appropriate sampling frequency, this initial PAT method is considered medium impact, and limited validation and model maintenance requirements are required.

INDIRECT PAT APPROACHES: SOFT SENSORS AND DIGITAL TWINS

The previous sections focused on direct PAT approaches; however, many products are too complex to measure all CQAs in real time. In many cases, a process variable or a CQA is considered hardly measurable because the measurement is not specific enough, the resolution does not suffice, or the variable is intracellular (in the case of biologics). In these situations, indirect measurement and correlation techniques are needed, and soft sensors, which use a combination of other measurements and mathematical calibration algorithms to accurately predict potentially difficult to measure variables, can be used.

There are multiple approaches for the generation of soft sensors [20]. The most desirable soft sensors are fully mechanistic ones, which use first-principle approaches, such as energy, mass, and elemental balances [21]. When complexity must increase to accurately predict the target variable, mechanistic modeling approaches are appended with data-driven correlations (also called hybrid models) [22].

Mechanistic or hybrid kinetic models can be implemented in a real-time environment and thereby mimic the process as a digital twin. Digital twins can be used to estimate the current state of process variables by using observed algorithms, such as Kalman or particle filters. Furthermore, they allow prediction and

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Overall, the quality requirements—and, thereby, the quality systems—are not changed by the implementation of continuous manufacturing.

(model-predictive) process control, a key requirement for continuous bioprocessing [23]. Because digital twins derive variables that cannot be easily measured, they can even be used as tools for controlling on variables that are normally not measurable. Hence, together with workflows known as “good modeling practice,” digital twins provide means to measure less and allow new opportunities for extended experimental design and enhanced process characterization [24]. Although these techniques are good to predict performance, they have limited use for establishing release criteria.

METHOD AND MODEL UPDATES

PAT methods focused on product life-cycle management (ICH Q12) require routine verification of performance beyond what is required for traditional test methods. Enhanced attention must be given to the validation of the models, as well as the continued verification that the model is operating in its validated space. Therefore, computational model life-cycle management, which includes routines for model diagnosis and drift detection, must be established [25]. This will enable identification of a transition point where models are allowed to self-learn on additional data sets, or where the model needs to be maintained by process experts. More research is required on the following questions: What would be necessary to allow a self-tuning model? What would preclude a self-tuning model and would require a process expert's intervention? How would life-cycle management be performed?

Q12 is proposed as the update mechanism for changes to all models, such as addition of spectra to a model or providing proof of being able to use a model for RTRT. Many strategies for self-learning algorithms have been in place for many years and will be enhanced by machine learning and artificial intelligence strategies in the future. However, the changes to a digital twin and soft sensor must remain traceable in a GxP environment. Moreover, the validated space of the method/model must be declared, and data integrity must be ensured, which can be a challenging task for data-driven approaches and will be a research field in the near future. We need to address questions such as the following: Do sufficient criteria exist to declare the validated space and ensure data integrity? If the validated space changes from a previously approved model but

is irrelevant for the operating space, is this a model change? In many cases, mechanistic approaches are encouraged because model parameters have a mechanistic meaning. Their value can be compared to literature values, which allows easier validation, and can therefore be easier to interpret and judge for model validity during unforeseen perturbations.

Another potential solution is to allow for the use of “or equivalent” with PAT methods, similar to what is done with traditional analytical methods, and only the spectra that were part of the original validation would be used for these equivalency assessments. To make this approach possible, updates to current regulatory guidance documents would be required, as described in the PAT validation section. If updates to guidance documents to allow for “or equivalent” can be realized, certain activities, such as the addition of a second software and the addition of a second analyzer of the exact same model with no change in original acceptance criteria, could be captured within the quality system. This would simplify reporting because changes could be submitted as part of the annual report instead of in a regulatory filing.

OTHER POTENTIAL HURDLES

When PAT is used in a CM process, the number of data points or measurements will be significantly higher than with a traditional batch process. This greater amount of data allows for the use of alternative statistical methodologies to show product quality, but it also requires additional discussions and strategies around sampling plans, as discussed in detail by De Los Santos and colleagues [26]. Overall, the quality requirements—and, thereby, the quality systems—are not changed by the implementation of CM. However, timing and responsibilities for activities such as batch review, release strategy, stability strategy, rejection strategy, and contingency plans for PAT may change.

As discussed previously, in CM, some activities may shift to an earlier phase in the manufacturing process development, especially because the process may be developed at scale. Similarly, process validation may also be done sooner because a CM line is more integrated than traditional batch processes (as discussed by the FDA in new draft guidance for CM [2]). The combination of PAT with CM may enable continuous process verification because it allows for immediate feedback of process performance and the use of a QbD approach if used throughout development.

CONTINGENCY PLAN FOR PAT

As described in guidance documents [17, 18], regulators expect that organizations will include a contingency plan as part of the strategy when implementing PAT. A range of potential approaches can be taken with respect to the contingency plan, and each has its own advantages and disadvantages.

Multiple PAT

One option for contingency planning involves creating a line with multiple equivalent PAT tools for a specific measurement. This can be done by ensuring available backup PAT tools to insert into the

line when the “primary” tool goes down, or the organization can have multiple tools built into the line at equivalent points to measure the same parameter.

It is critical that organizations have a predetermined, scientific and risk-based strategy around hierarchy of the multiple values/PAT outputs during routine operation to eliminate confusion. The backup PAT tool can also be offline equipment not intended for direct connection to the line.

Alternative Test Methods

If PAT is unavailable, or does not function as expected (i.e., does not pass suitability tests), alternative or traditional test methods can be employed. It should be noted that this statement only applies to scenarios where the testing method has been appropriately documented through the control strategy and overall product risk. One drawback to this approach is the organization may need to collect a high volume of samples to show process control. Also, the traditional test usually takes longer than the PAT test and can delay product release, especially if RTRT is employed.

Shutdown

If an organization uses PAT but does not use multiple PAT tools or an alternative test method as the contingency plan, it may need to shut down the line if the PAT is unavailable. In this scenario, no product is manufactured until the PAT tool is again available. This will require the organization to maintain higher inventory volumes to minimize the risk for potential out-of-stock events, which can be cost prohibitive. The benefits, although minor, of this approach are that the organization will not require additional PAT tools or analytical equipment as part of its upfront investment and long-term maintenance costs may be lower.

If a company is not willing to invest in backup PAT, they are also less likely to implement PAT from the start due to the high risk of not being able to run the standard planned commercial process or meet the demand plan. This can then lead to drug shortages, as noted previously, which is otherwise a problem that CM can alleviate.

CONCLUSION

Throughout this article, we have shown how PAT can aid in the implementation of CM throughout the life cycle and have highlighted areas of concern for PAT implementation. Based on the product and company strategy, PAT can be used for process development, control, monitoring, and product release. Guidance documents are clear on the requirements and expectations for PAT when used for real-time release or to make final quality decisions. However, there are major hurdles for increased implementation of PAT within CM, for which we have proposed some solutions:

- Validation requirements for medium-impact models.
- Model maintenance and updates: Guidance documents from various health authorities offer differing requirements regarding the expectations for model maintenance. Our recommendation is to continue harmonization discussions as

Q13 and Q14 are being drafted to clearly capture what is expected and to align expectations with requirements for traditional test methods. Also, by shifting to a performance-versus-parameter-based description of methods, more changes can be managed through the PQS instead of in regulatory filings. This would decrease the time requirement for implementation of a change and allow for increased use of PAT.

- Processes and system for traceability: The use of PAT requires models, which need to be adjusted throughout the product life cycle. Therefore, it is critical to have tools to trace the need to adapt the models in a GxP environment. Workflows and decision trees must be available as data science solutions.
- Skills, knowledge, and mindset within the organization: It is important to share knowledge and develop the skill set of the entire organization from the introduction of PAT and CM to allow for better understanding and sustainability of these initiatives.
- Business case requirements: Companies need to think holistically about how they develop the business requirements for and benefits of PAT. The first application will always be relatively costly and time consuming; however, once the tool and the skills are in place, companies can realize significant benefits at significantly lower costs. Through the use of PAT, companies may also increase overall knowledge of the process, which supports the holistic approach to the business case.
- Options when unable to measure all CQAs with PAT: Companies should acknowledge these issues and develop appropriate processes for modeling if PAT tools are unavailable.
- Software solutions: Workflows for digital twins are in place; however, a digital twin needs a suitable real-time environment, and standardization is key in this respect; therefore, the industry needs software as a service (SaaS) solutions. 

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